

INTRODUCTION TO COMPUTER SYSTEM, NETWORKS AND SECURITY

Assignment 3 part 1 – Networks (Reflection)

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Purpose for the whole assignment 2:

- ❖ In this assignment, students will record data from a real-world wireless network and demonstrate that they can analyse it, identify its properties and potential issues. Also, students will analyse Internet traffic and identify servers, clients and protocols used. The assignment is related to Unit Learning Outcomes 5 and 6.

Purpose for the assignment 3 part 2:

TASK	BRIEF DESCRIPTION
Reflection	Reflect on what was done during week 7 to week 9 of classes; what you learned, what was difficult...

FIT1047 Weekly Reflection

Week 7:- Networks: physical, data link, network and transport layers

Key Learnings:

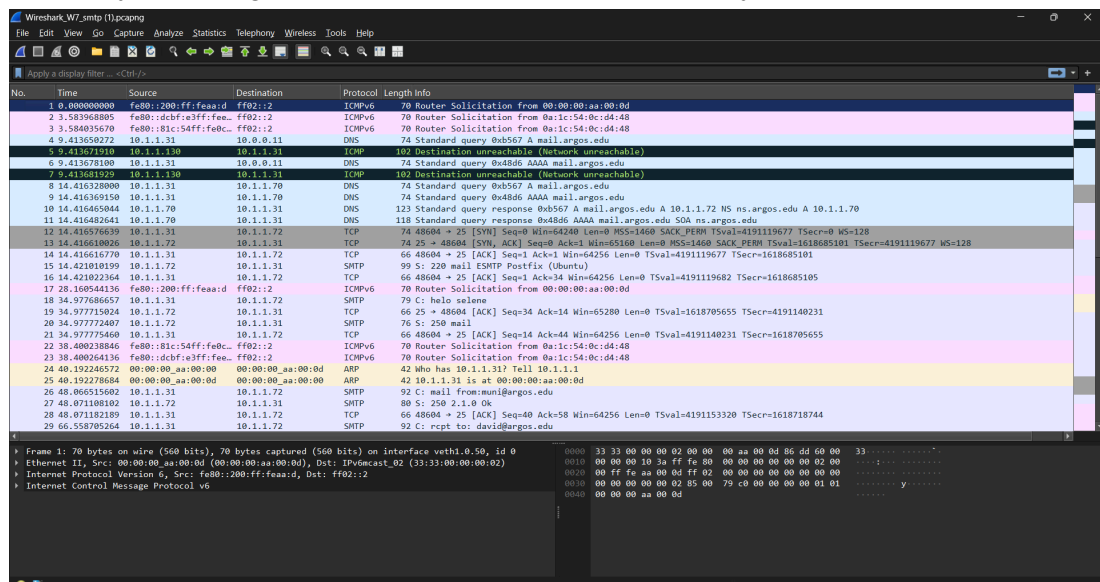
- Describe different ways of encoding digital data into digital and analog signals
- Identify different network interface cards and cables
- Understand contention-based media access control at the Data Link Layer
- Understand the difference between IP versions 4 and 6
- Describe how the network layer routes packets (routing tables and dynamic routing)
- Understand the difference between UDP (connectionless) and TCP (connection-oriented) transport layer protocols
- Distinguish between addressing at the different layers

Personal Insights:

- Learning about how the data is transmitted over the network was very interesting

Application of Knowledge:

- I used my knowledge learned in the applied class to analyse data in wireshark.



Challenges Faced:

- Getting used to wireshark was a bit difficult at first but I quickly got used to it.

Future Goals:

- Continue to practice wireshark to gain practical knowledge and do research to understand theoretical concepts

—Assignment 3—

Networks

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Week 8:- Networks: LANs & WANs

Key Learnings:

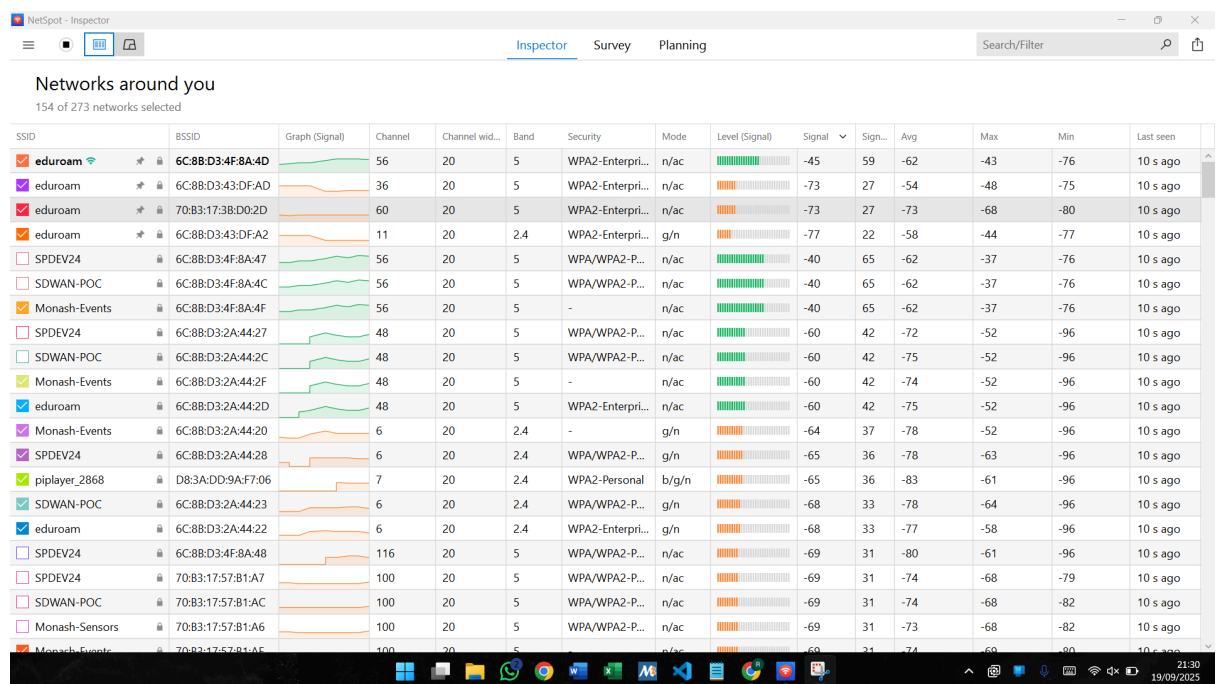
- Understand Media Access Control for Ethernet and WLAN
- Learn about WLAN (service sets, channels, typical set-up)
- Describe how forwarding tables work in Ethernet switches
- Understand how large organisations can organise their networks in LANs, WLANs and WANs
- Learned how to use a wifi analyser(NetSpot)

Personal Insights:

- Learning how networks are created and are used was very interesting.

Application of Knowledge:

- I used my knowledge learned in applied class to answer the question
- I also used that knowledge for the assignment 3



Challenges Faced:

- Understanding how to use netspot was tricky at first but I got used to it very quickly.

Future Goals:

- Review the workshop and applied class to get ready for the assignment.

—Assignment 3—
Networks

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Week 9:- The Internet

Key Learnings:

- Describe the structure of the Internet (Autonomous Systems, IXPs, CDNs)
- Discuss strengths of the design (decentralised, based on open standards)
- Discuss weaknesses of the design (threats, control)
- Explore possible future directions for the Internet

Personal Insights:

- Learning how the internet works was very interesting.

Application of Knowledge:

- I used my knowledge in applied class to do the question/activities.

Challenges Faced:

- Learning about Digital Transmission NRZ-Invert and Manchester was a little difficult for me but after doing the activity in the workshop class, I was able to grasp the concept.

Future Goals:

- To do personal research and deeper understanding about the internet.
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